Aggregate transcript projectors

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The destination extractor now multiplies exact sign indicators directly in the split function algebra F_r[X]/(X^r−X). It avoids enumerating scalar-field values for compiler synthesis and avoids an explicit subset expansion. It is connected to the standalone public-point sign source with mandatory final group verification.

Public Q+iP ⟶ standalone sign s ⟶ exact indicator E(a,s)

Update I ⟵ I·E mod (X^r − X)

Count N = −coeff[r−1] Moment M = −coeff[r−2]

N=1 ⟶ candidate M ⟶ verify [M]P = Q

No subset expansion; coefficient state still has dimension r

Construction

For C=(X+a)^((r−1)/2), the indicators are (C²+C)/2 for sign+1, (C²−C)/2 for sign−1, and 1−C² for zero. Their product is exactly the candidate-set indicator, including every zero endpoint. Two coefficients give candidate count and first moment. After one constraint the count is below r, so its residue is the exact integer count. A singleton's first moment is its scalar. The empty transcript is explicitly excluded from this count rule.

Software checks

Three selected native methods pass, with 157 imported methods filtered. There are 324 transcript-prefix checks over scalar primes 5, 7, 11 and 31, verifying count, moment, every function value and idempotence against a test-only enumerated reference. Repeats, contradictions, zero endpoints, caps and invalid inputs are checked. All 20 frozen public-point cases at p=509, r=479 recover and pass forward verification; the one-query control remains unresolved. No original-cohort rerun or timing panel.

Twenty-six dependency hashes are pinned. Normal dep-info excludes legacy source/transport paths and quartic/bitplane compiler modules. Test and normal builds retain 16 and 28 warnings. No build or selected test failed. Checks and proof are producer artifacts, not independent review.

Cost and next construction

Each update requires O(M(r)+r) scalar-field work and retains a coefficient state of dimension r, plus multiplication temporaries. A public inverse-coefficient array costs O(r) setup. Source queries retain their existing cost/domain restrictions; this adapter does not yet use shared windows.

A nonempty proper candidate indicator necessarily has degree r−1 because its leading coefficient is minus its nonzero count. Degree trimming cannot compress it. Its other coefficients encode candidate power sums; next investigate conversion to a low-degree root polynomial once the count falls, charging conversion and subsequent updates explicitly. This is a representation constraint, not a general circuit lower bound.

Sources and replay

DERIVATION.md gives identities, costs and the validation command context; run validate.py with a new output directory. Existing quotient constructor and CONTINUATION.md guided the approach; KB and the older lifted-moment report were unavailable. Primary sources checked: Sage quotient-ring documentation and van Dam–Hallgren–Ip's shifted-character problem paper. No novelty claim or quantum algorithm.